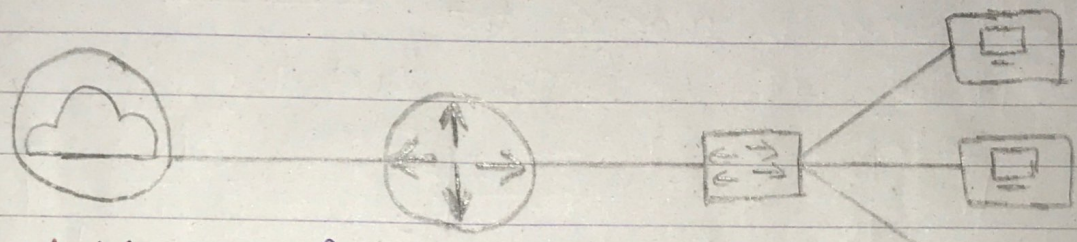
21 Sep,

Lecture-1

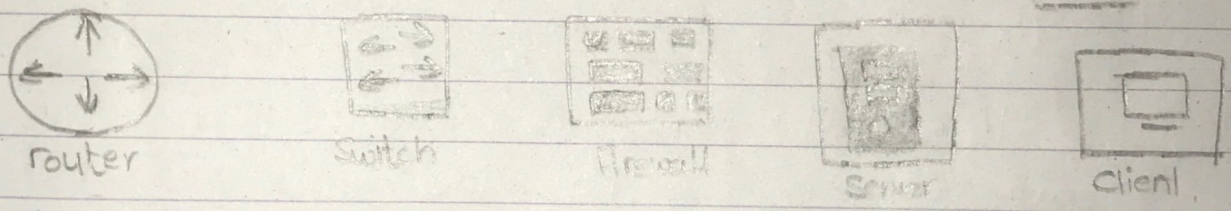
Network Devices.

What is computer network?

↳ A computer network is a digital telecommunication network which allows nodes to share resources.



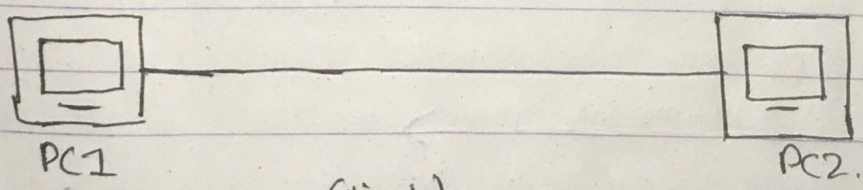
Now, what is a node?



All these are nodes

Like clients, as well as other node type, servers are also referred to as end hosts or end points.

~~Client car~~



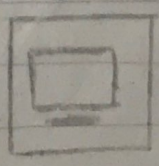
[Two PCs, we connect them with wire. Now they're in the network]

(clients)

These, two nodes, are connected so, they can share resources. They and have communication with each other.

Client.

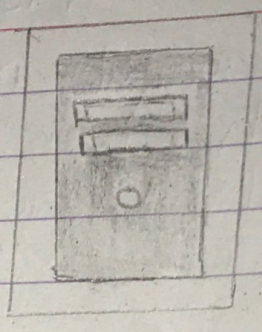
↳ A client is a device that ~~acce~~ accesses a service made available by a server.



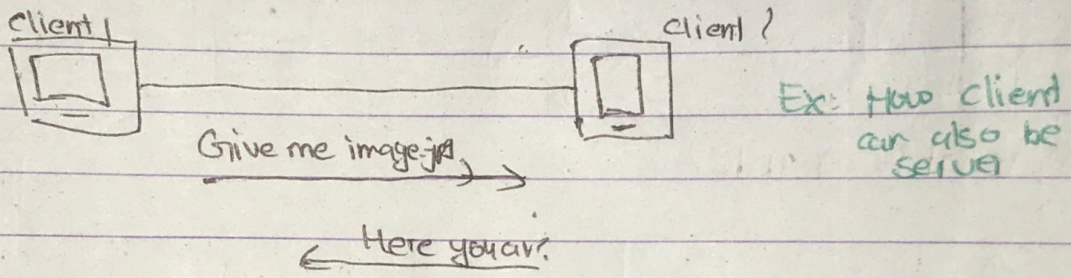
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Server.

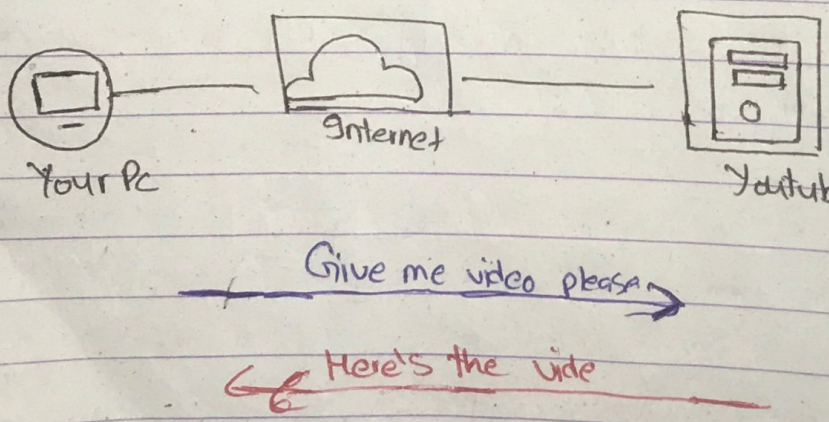
(Servers are also in data center)
Client can also be server.



↳ A server is a device that provides services or functions for clients.



Here client 1 request for service & client 2 provides the service (image.jpg). So client 1 is client & client 2 is server.



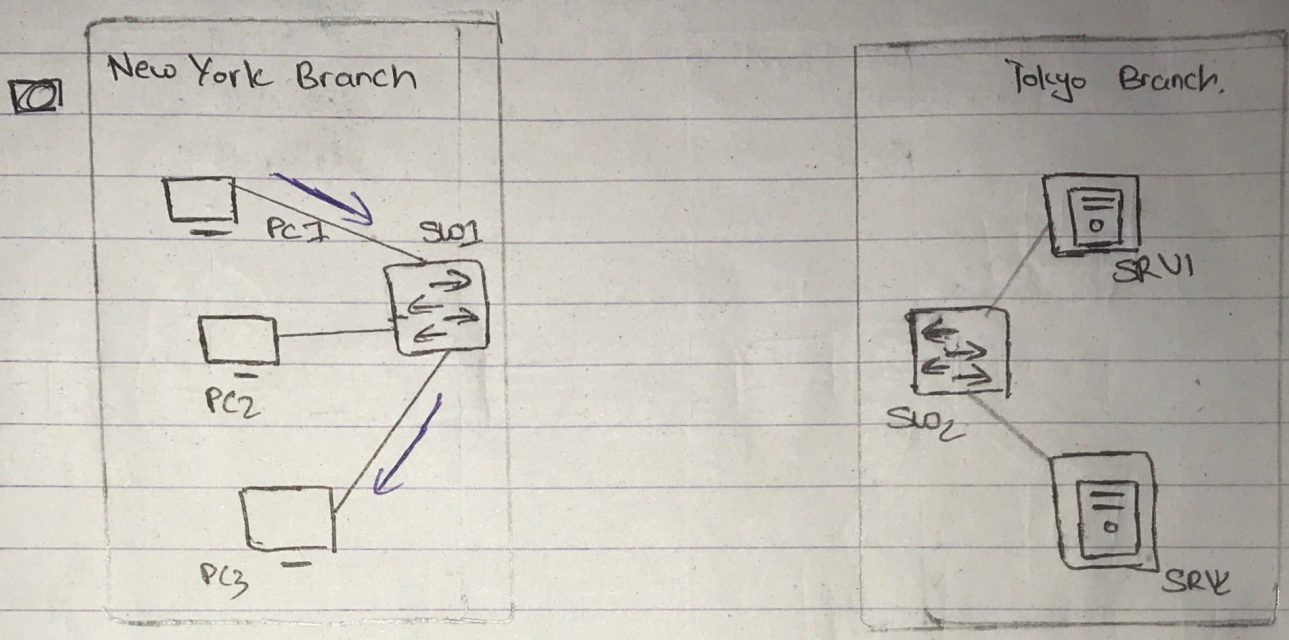
The same device can be client in some situations & server in other situations.

Server and client are known as endpoints/end hosts

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Switch - use designed to connect many hosts in the same LAN together.

Switches.



↳ Switches are used to forward traffic within a LAN. PC1, PC2, PC3 & SW1 plugged into switch1 all reside on same LAN. The same goes for SRV1, SRV2 are plugged into SW2 in Tokyo branch.

one

Now, we have a LAN on Newyork & other on Tokyo. The hosts within each LAN can send data to each other. Example PC1 to PC3.

These switches cannot connect to internet directly & send data between two LANS.

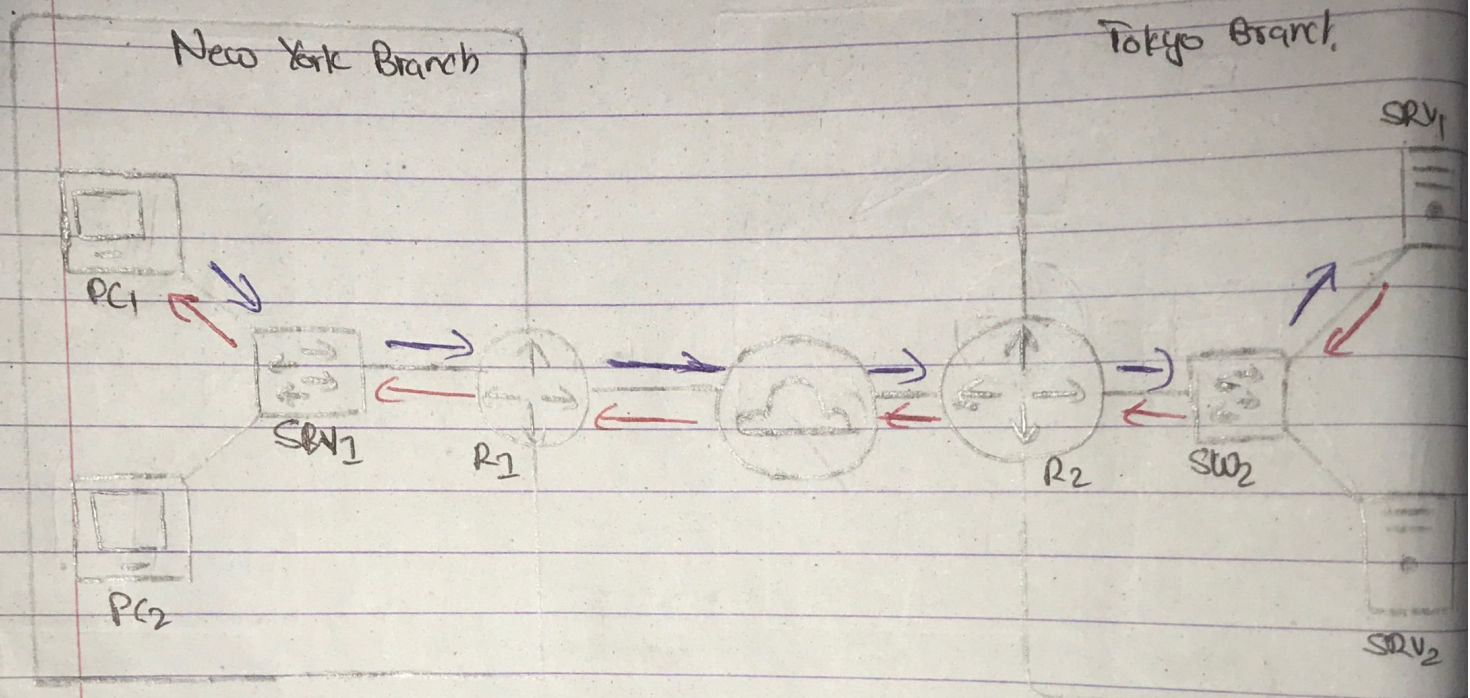
Characteristics of switches.

- they have many network interfaces/ports for end hosts to connect to (usually 24+).
- the they provide connectivity to hosts within the same LAN.
- they do not provide connectivity between LANs or over the internet.

(Catalyst 9200, Catalyst 350 are some example of switches)

A router is designed for forwarding traffic between networks. It can't connect lots of hosts like PCs.

Routers.



When endhost in NY LAN want to communicate with endhost on Tokyo Tokyo LAN, they will send their data to router, R1, which will then forward it to the Tokyo Branch LAN via internet.

If PC1 wants file from SRV1, it will send the req to R1 via SW1, which will then forward it over the internet to R2, which then send it to SRV1 via SW2. and the reply will follow reverse path to PC1.

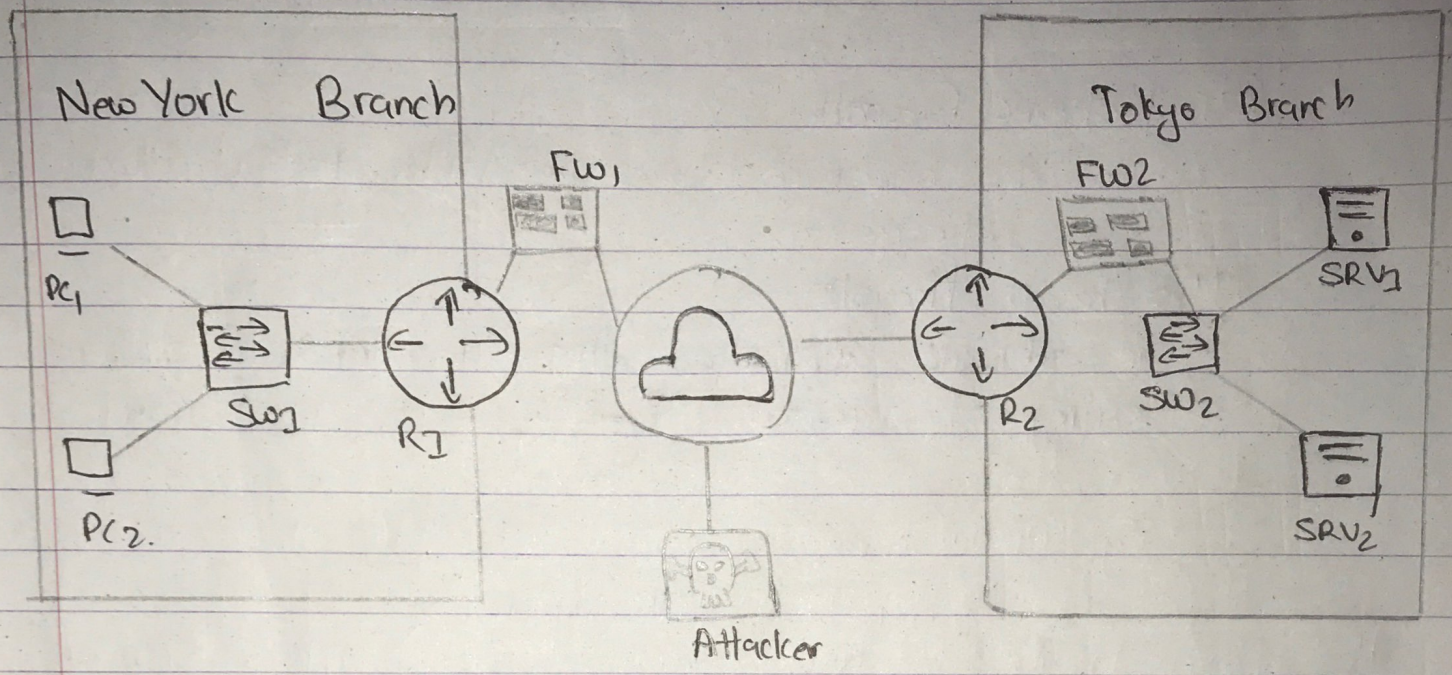
Characteristics of Router

- ISR 1000, ISR 900, ISR 4000 are some eg. of CISCO enterprise routers
  ISR 1000 & 4000 have network interfaces on back. while 900 have on front
- Routers have few network interfaces (typ device with Port/hole) in comparison to switch.
- Routers are used to provide connectivity between LANS because of that Routers are used to send data over internet.

ASA5500X
Cmodesr!

ASA – Adaptive Security Appliance
(classic firewall).
Firepower 2100 – Next gen firewall
consists of IPS

Firewalls (Firewalls can be physical device as well as software)



Routers provide some basic security features, we should actually use firewalls to protect our network.

↳ Firewall are specialty network security devices that control network traffic entering and exiting your network. They can be placed outside of Router like FW1 or inside of router like FW2.

What's important is that they protect the end hosts inside, like PCs and Server in this network.

Firewalls must be configured with security rules to determine which network traffic should be allowed & which should be denied defined

Some Cisco firewalls: ASA5500-X, Firepower 2100
Modern firewall include IPS (Intrusion Prevention System)

Some characteristics of firewall

- they monitor and control network traffic based on configured configuration rules.
- can be placed inside or outside network (can filter traffic before or after it reaches router)
- are known as 'Next-Gen' when they include modern and advance techniques.
- they can In some networks, they are placed both inside & outside (ie 2 firewalls)

Firewall on our computer.

Network firewall.
- are hardware devices that filter traffic between networks.

Host-based firewalls
- are software applications that filter traffic entering & exiting a host machine, like PC

- Eg: Windows defender

~~Even in network firewall, there should be software firewall for extra line of defense.~~ network with hardware

Even in network & with hardware firewall, each PC should also include software firewall for extra line of defense.

Quiz:

Q1) # Your company wants to purchase some network hardware to which they can plug 30 pcs in your depart. Which type of network device is appropriate?
A) Router B) Firewall C) A switch d) A server

Q.2) Yar received a video file from your friend's apple iphone using airdrop what does his iphone functioning as in that transaction?
A) Server B) Client C) LAN

Q3) What is your Computer or mobile functioning as we watch this video? (YT)
A) Server b) An end host C) Client

Q.4). Your company wants to purchase some network hardware to connect its separate networks together. What kind of network device is appropriate?

A) Firewall B) A host C) A LAN d) Router

Q.5). Your company wants to upgrade its old network firewall that has been in use for several years to one that provides more advanced functions. What kind of firewall shall they purchase.

a) Host-based b) next-level c) next-generation d) top-layer

Next-level & top-layer are not actual types of firewalls

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